

Exam_1_3

November 20, 2024

```
[ ]: import numpy as np
np.set_printoptions(precision=1)
def phi(i):
    M=np.array([[1,0,.5],[0,1,.5],[.5,.5,1]])
    eigenvalues,eigenvectors=np.linalg.eig(M)

    return np.array([0, np.sqrt(eigenvalues[0])*eigenvectors[i,0],np.
↪sqrt(eigenvalues[1]))\
        *eigenvectors[i,1],np.sqrt(eigenvalues[2])*eigenvectors[i,2]])
Phi=np.zeros([1,4]),*[phi(i) for i in range(3)]
K=np.zeros([4,4])
for i in range(4):
    for j in range(4):
        K[i,j]=np.round(np.dot(np.array(Phi[i]),np.array(Phi[j]).T),4)
```

```
[ ]: print(Phi)
```

```
[array([0., 0., 0., 0.]), array([ 0. , -0.7, -0.7, -0.3]), array([ 0. , -0.7,
0.7, -0.3]), array([ 0.0e+00, -9.2e-01, -4.1e-16,  3.8e-01])]
```

```
[ ]: print(K)
```

```
[[ 0.  0.  0.  0. ]
 [ 0.  1. -0.  0.5]
 [ 0. -0.  1.  0.5]
 [ 0.  0.5 0.5  1. ]]
```